

GOWRISHANKAR S

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PROFESSIONAL SUMMARY

AI Developer with hands-on expertise in Generative AI, Large Language Models (LLMs), and traditional Machine Learning. Experienced in building production-ready LLM-powered applications using LangChain, LangGraph, and RAG architectures, alongside multi-agent orchestration workflows using MCP and A2A protocols. Proficient in Python, SQL-based ETL pipelines, and data preprocessing. Strong background in Computer Vision, Edge AI, and Multimodal systems, with a proven track record of shipping real-time, production-grade AI products across manufacturing, accessibility, and research domains.

TECHNICAL SKILLS

Generative AI & LLMs: GPT Models, LangChain, LangGraph, RAG Architectures, Prompt Engineering, Vector Databases, Multi-Agent Orchestration, MCP Protocols, Transformers, NLP

Traditional ML & Deep Learning: Classification, Clustering, Regression Analysis, Decision Trees, Neural Networks, PyTorch, TensorFlow, Scikit-learn, Predictive Modelling, Computer Vision (YOLOv8, OpenCV)

Engineering & Cloud: Python, SQL, ETL Design Patterns, FastAPI, Flask, Docker, CI/CD Pipelines, Git/GitHub, Streamlit, Linux, RESTful APIs, Node.js

Data, Hardware & Mobile: Data Preprocessing & Transformation, Power BI, Data Visualization, Google Sheets API, Raspberry Pi, IoT Sensor Integration, Android, AR

PROFESSIONAL EXPERIENCE

Forge Innovation and Ventures July 2025 – Present, Coimbatore, Tamil Nadu

Forge Fellow

- Designing and deploying production-ready AI/ML solutions including LLM-powered pipelines, RAG systems, and multi-agent workflows for real-time, low-latency environments.
- Building end-to-end data pipelines leveraging SQL for ETL, preprocessing, and transformation to feed both GenAI and classical ML models.
- Implementing multi-agent orchestration architectures using LangGraph and MCP/A2A protocols for complex, autonomous task execution.
- Leading full-lifecycle AI product development — from data acquisition and model training to REST API integration and cloud deployment.
- Enforcing MLOps best practices including Docker containerization, CI/CD pipelines, and model versioning for robust, repeatable deployments.

EDUCATION

B.Tech, Artificial Intelligence & Data Science

Kumaraguru College Technology - Coimbatore

2022-2025 CGPA: 7.5

Diploma in Computer Science Engineering

Sri Ramakrishna Polytechnic College - Coimbatore

2019-2022 77%

KEY PROJECTS

Visual Inspection System for Manufacturing

Computer Vision · Complete

- Led design and development of the CV pipeline, building algorithms to detect object dimensions and validate components against digital blueprints.
- Owned the hardware-software integration, engineering the plug-and-play conveyor device for real-time defect detection on the line.
- Drove automation of the validation workflow end-to-end, directly reducing manual inspection errors and increasing throughput.

YOLOv8 · OpenCV · Python · Hardware Integration · Real-Time CV

Object Navigation System for Visually Impaired

Edge AI + Computer Vision · Complete

- Optimized and deployed YOLO architectures on Raspberry Pi 4 using quantization, owning edge-inference performance end-to-end.

- Designed and built the multi-modal (Audio + Haptic) feedback system from the ground up for spatial awareness.
- Architected the fail-safe navigation logic, prioritizing obstacle avoidance in dynamic indoor environments.

YOLO · Raspberry Pi · Python · Computer Vision · Edge AI · Audio/Haptic

AI Assistant (Multimodal)

Multimodal AI · Deployed

- Owned integration of speech recognition, computer vision, and NLP modules into a single interactive assistant.
- Designed and built modular pipelines for transcription, response generation, and voice synthesis.
- Handled real-time vision input processing, ensuring low-latency responses across all modalities.

NLP · AssemblyAI · ElevenLabs · PyTorch · YOLOv8 · Python

AI-Powered Electronic Component Detector

AI Web App · Live

- Built the end-to-end detection system using Transformers and PyTorch to identify and classify PCB components.
- Developed the Flask web interface, enabling batch uploads, live camera streams, and confidence-based filtering.
- Optimized model inference for real-time video processing, delivering instant feedback to assembly-line operators.

Deep Learning · Flask · Transformers · OpenCV · PyTorch

Real-Time Energy Monitoring Dashboard

Dashboard · Complete

- Built the Streamlit dashboard end-to-end, connecting it to Google Sheets for live energy telemetry.
- Implemented automated polling and alert-threshold logic to flag anomalies in real time.
- Owned performance tuning for low-latency live updates suitable for production environments.

Python · Streamlit · Google Sheets API · Data Visualization

Automated Agriculture Drone — Disease Identification

Research AI · Complete

- Collaborated directly with agricultural researchers to fine-tune drone-based disease identification models.
- Built the computer vision layer-tracking pipeline used to evaluate model accuracy in the field.
- Validated results against manual inspection, achieving an 80% improvement in pathogen detection.

AI · Drone · Computer Vision · Python

Indoor Navigation Using AR

Mobile AR · Deployed

- Developed the Android app end-to-end, integrating AR technology for indoor wayfinding.
- Mapped and calibrated the navigation model specifically for the college campus environment.
- Measured and validated impact, contributing to a 75% reduction in student navigation tardiness.

Android · AR · Mobile UX

SwiftSports — Ecommerce Platform

E-Commerce · Prototype

- Designed and built the ecommerce prototype end-to-end, covering product listing through checkout.
- Implemented cart management and product flow logic using JavaScript and Node.js.
- Optimized the checkout experience to maximize conversion within the prototype.

HTML/CSS · JavaScript · Node.js

Smart Parking System

IoT + Mobile · Prototype

- Built the mobile interface for real-time visualization of parking slot status.
- Integrated IoT sensors to capture and transmit live parking availability data.
- Designed the real-time monitoring logic connecting sensor data to the app interface.

IoT · Mobile App · Real-Time Monitoring